Homework 4 – STAT 6611

Due Friday, November 13 by midnight (submit in gradescope under 'Assignments')

1. Let $\underline{x}$ be an $n \times 1$ random vector with mean vector $\underline{\mu}$ and variance-covariance matrix Σ. Use Theorem 3.1 to show that $\underline{x}$ has a multivariate normal distribution if and only if, for every $n \times 1$ vector of constants $\underline{a}$, $\underline{a}'\underline{x}$ has a (univariate) normal distribution.

2. Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$. Show that

$$\text{cov}(\underline{x}' A \underline{x},\ \underline{x}' B \underline{x}) = 2\ \text{tr}(A\Sigma B\Sigma) + 4\underline{\mu}' A\Sigma B \underline{\mu},$$

where A and B are symmetric matrices of constants. [Hint: For any random variables w and z, $\text{Var}(w+z) = \text{Var}(w) + \text{Var}(z) + 2\,\text{cov}(w,z)$.]

3. Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is an $n \times 1$ vector and rank $(\Sigma) \geq 1$. Let

$$w_1 = \underline{b}_1' \underline{x} + \underline{x}' A_1 \underline{x} \ \text{ and } \ w_2 = \underline{b}_2' \underline{x} + \underline{x}' A_2 \underline{x},$$

where $\underline{b}_1$ and $\underline{b}_2$ are vectors of constants and A_1 and A_2 are symmetric matrices of constants. Let

$$\underline{c}_1 = \underline{b}_1 + 2A_1\underline{\mu} \ \text{ and } \ \underline{c}_2 = \underline{b}_2 + 2A_2\underline{\mu}.$$

   (a) Find

$$\text{cov}\left(\begin{bmatrix} \underline{b}_1' \\ A_1 \end{bmatrix} \underline{x},\ \begin{bmatrix} \underline{b}_2' \\ A_2 \end{bmatrix} \underline{x} \right).$$

   (b) Suppose that (i) $\underline{b}_1'\Sigma\underline{b}_2 = 0$, (ii) $\underline{b}_1'\Sigma A_2 = \underline{0}'$, (iii) $\underline{b}_2'\Sigma A_1 = \underline{0}'$, and (iv) $A_1\Sigma A_2 = 0$. Use (a) to show that w_1 and w_2 are independent. [Hint: Note that $w_i = \underline{b}_i'\underline{x} + (A_i\underline{x})' A_i^{-} (A_i\underline{x})$ for $i = 1, 2$.]

4. Take the model to be Model II and assume that $\underline{y}$ has a multivariate normal distribution.

   (a) Obtain the expectation, the variance, and the mean squared error for any estimator of σ^2 having the form $(1/k)\underline{y}'(\underline{y} - X\widehat{\underline{\beta}})$, where k is a nonzero constant, and $\widehat{\underline{\beta}}$ is any solution to the normal equations.

   (b) Of those estimators of σ^2 having the above form, which has the smallest mean squared error? (That is, determine the optimum value of k.)

5. Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is an $r \times 1$ vector and rank $(\Sigma) = k \geq 1$ (where possibly $k < r$). Let A be an $r \times r$ symmetric matrix. Show that, if $A\Sigma$ is idempotent of rank s and if in addition $\underline{\mu}$ lies in $\mathcal{C}(\Sigma)$, then $\underline{x}' A \underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}' A \underline{\mu}\right)$. (Hint: Let Γ be a $k \times r$ matrix such that $\Sigma = \Gamma'\Gamma$, and show that $A\Sigma$ is idempotent of rank $s \Longrightarrow \Gamma A \Gamma'$ is idempotent of rank s. Then, use $\underline{x}$ has the same distribution as $\Gamma'(\underline{z} + \underline{a})$, where $\underline{z}$ is a standard normal vector and $\underline{a}$ is a certain vector of constants.)

6. Let $\underline{x} \sim N(\underline{0}, I)$, and take A and B to be two symmetric matrices such that $\underline{x}' A \underline{x} \sim \chi^2(p)$ and $\underline{x}' B \underline{x} \sim \chi^2(q)$, where $p > q$. Show, by completing the following three steps, that $\underline{x}'(A - B)\underline{x} \sim \chi^2(p-q)$ if and only if the correlation corr $(\underline{x}' A \underline{x}, \underline{x}' B \underline{x}) = q/\sqrt{pq}$.

   (a) Show that tr $[(I - A)B] = 0$ if and only if corr $(\underline{x}' A \underline{x}, \underline{x}' B \underline{x}) = q/\sqrt{pq}$. [Hint: Consider cov $(\underline{x}'(A - B)\underline{x} + \underline{x}' B \underline{x}, \underline{x}' B \underline{x})$]

   (b) Show that $(I - A)B = 0$ if and only if tr $[(I - A)B] = 0$.

   (c) Show that $\underline{x}'(A - B)\underline{x} \sim \chi^2(p-q)$ if and only if $(I - A)B = 0$.